

Sociolinguistic Mobility: A Lens for LLM Pretraining Data Curation

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Motivations

- Data curation algorithms for large language model (LLM) pretraining are designed to mine “quality” from the web – a murky notion at best
- Quality filters (ex: HuggingFace’s FineWeb) optimize for benchmark performance as a proxy for “general” capabilities to model languages such as English
- Yet the widespread adoption of English on (and off) the internet is not uniform; it inherits from past and ongoing processes of migration, contact and standardization

Theoretical Framework

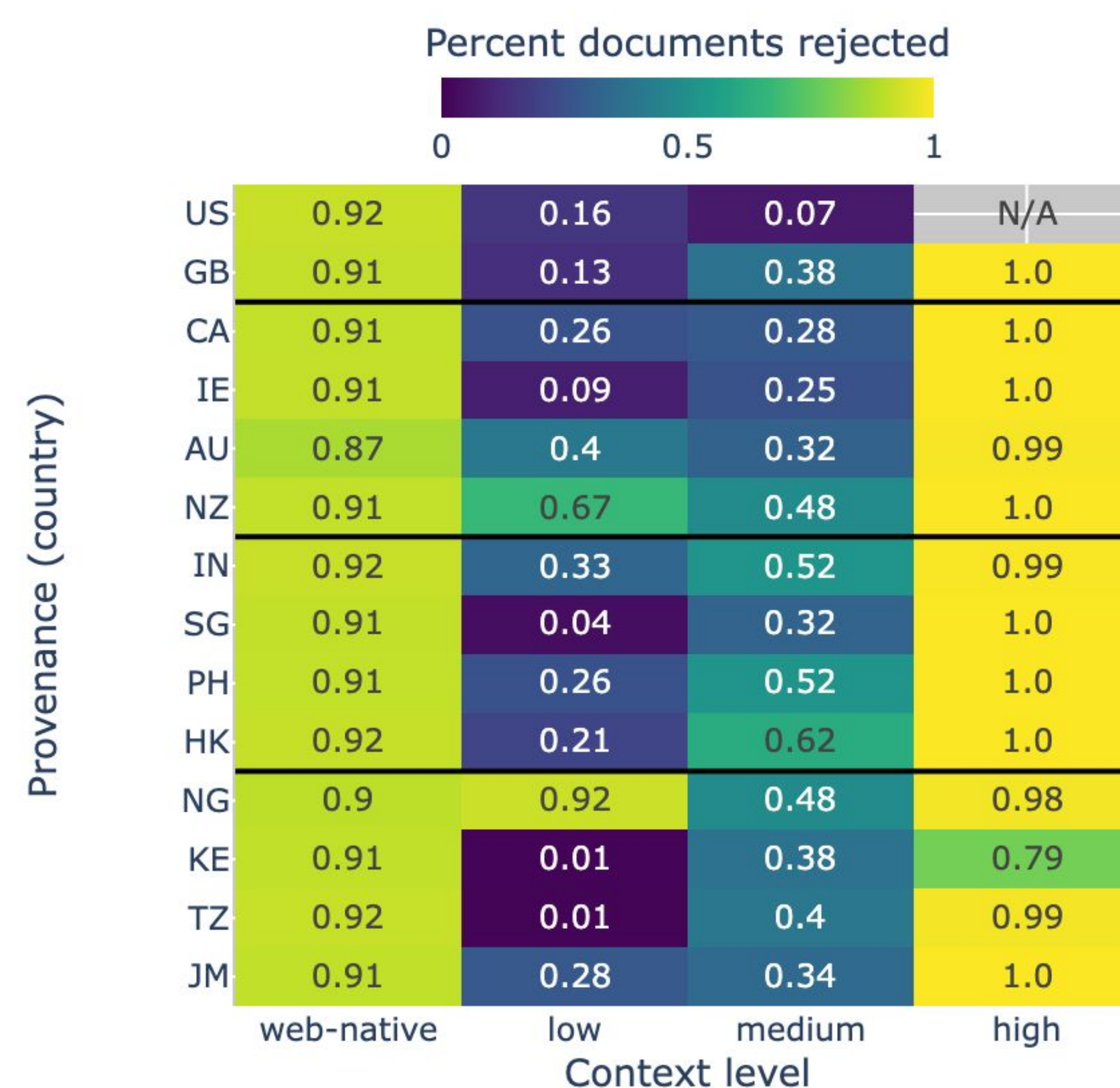
- Blommaert 2010 proposes a theory of sociolinguistic mobility: variation in language can be attributed by the mobility of people through sociocultural and spatiotemporal spaces
- This reflects the multi-dimensional nature of language use in terms of (1) geographic provenance of authorship and (2) the level of shared context between author/audience
- Geographies are grouped by shared/similar history of English adoption (cf. Kortmann et al. 2020)
- Context levels are grouped by similar situational characteristics (cf. Biber & Conrad 2009)

What is deemed “good quality” text for LLMs?

Experiment

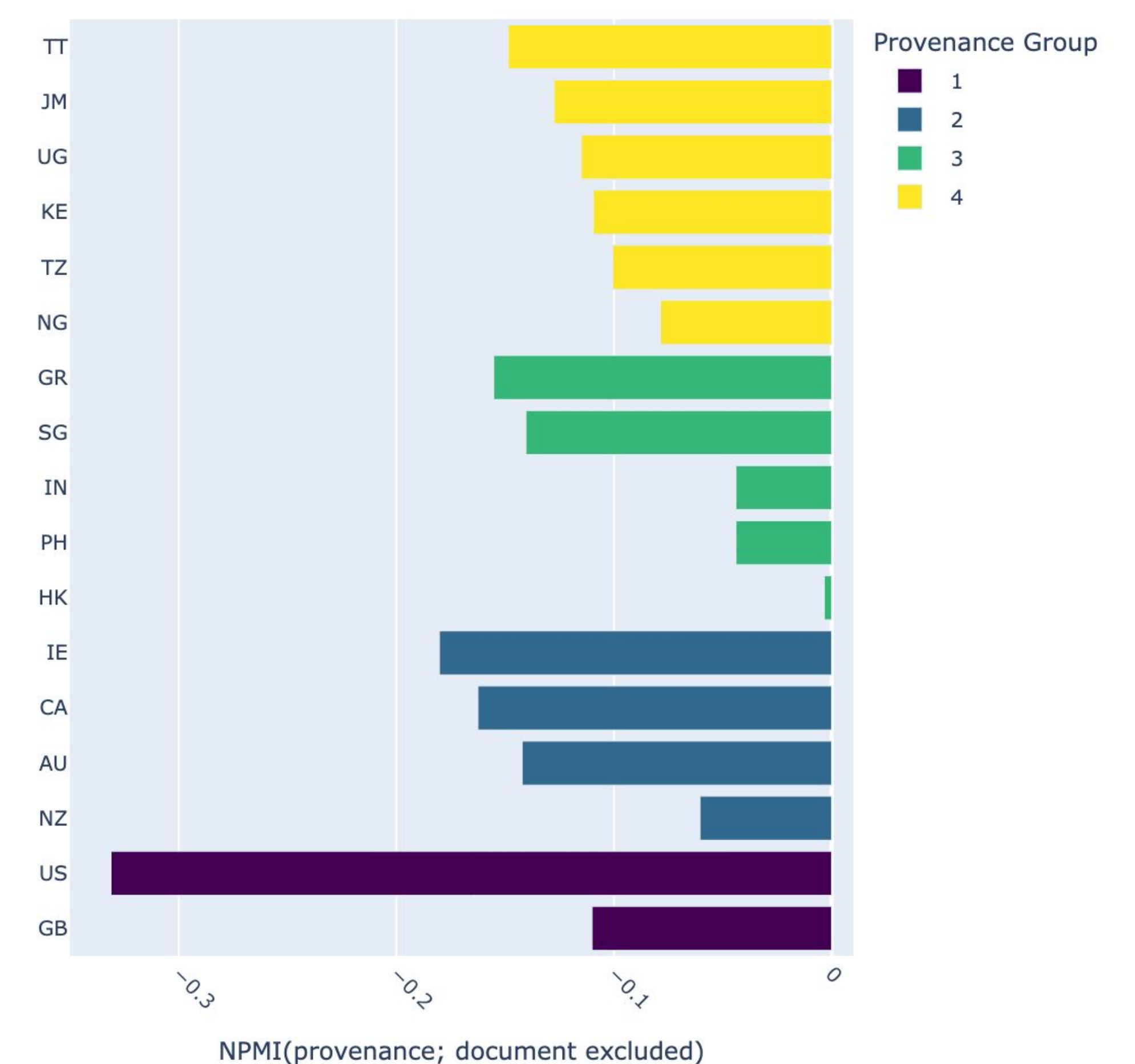
What effect does HuggingFace’s FineWeb filtering pipeline have on a diverse corpus of English language use?

Across low and medium context levels, United States authorship (n=200) makes documents ~7.8x less likely to be rejected. Within medium context, normalized pointwise mutual information (NPMI) suggests a statistically significant protective effect against rejection for US authorship. Meanwhile, FineWeb aggressively filters both web-native and high-context texts of all geographies under study.



Takeaways & Future Work

- Understanding documents as sociolinguistic objects can shed light on the nature of “quality” which LLM builders have not rigorously defined
- Pretraining data curation for “general” language modeling via benchmark performance is subjective and finite
- Future work will examine the manifestation of “quality” w.r.t. non-deterministic methods (e.g., data mixing)



Example Texts

✗ denotes FineWeb filter rejection, ✓ denotes FineWeb filter retention

About this item
As stated in the thesis project, "My community consists of over 650 non-profit organizations. These agencies range in size from the small local grassroots organization to fully staffed, fully funded government/private agencies. They consist of Soup Kitchens, Daycares and Senior Centers, M/R and M/H facilities, Drug/alcohol rehabs and Churches of all different denominations. The common thread that these agencies hold is the fact that they are all "social service agencies" that contain a feeding program that serves low-income individuals in some form whether through hot-prepared...

✗ Web-native, undetermined shared context (Tanzania)

Molecular biology has revolutionized the study of protein structure and function.
Not only has the microbial production of enzymes and other proteins in useful amounts become routine in recent years, but systematic alteration of protein sequence by site-directed mutagenesis is now a standard tool for dissecting chemical mechanism.
The recent development of new methods for ligating peptides, however, promises to make chemical synthesis of large proteins an attractive alternative to biosynthesis, particularly for the construction of novel molecules containing non-natural amino acids...

✓ Low level of shared context (United States)

Longer records, special jails, 'zebra' plates might be in future for offenders
If you are caught drinking and driving in the future, you might be sent to a special prison, be forced to attach a license plate to your vehicle marking you as a chronic drunk driver and have a record that will haunt you for a long time.

The Wisconsin affiliates of Mothers Against Drunk Driving, a national lobbying group with chapters in 48 states, hopes such legislation will be passed this year to help keep drunk drivers off the highway.
"One measure we would like to see passed is ...

✓ Medium level of shared context (United States)

Uh Li Po Chun United World College of Hong Kong
Uhm it 's kind of international school
Uhm we have
Oh uhm went into it
Who went to that school you mean
Uhm uhm
Uhm uhm
The school is not runned by government
I think uhm it uhm it it 's funded by the United World Organization
There 's There 's of totally eight United World College of in all the world...

✗ High level of shared context (Hong Kong)

<https://bit.ly/4jz6s0f>

Full paper,
references, source
code

